Coaching How To Search

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- Consider the following examples of AI agents:
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  - Agent B provides "study at home" support for students.
  - Agent C serves as a cognitive substitute of a therapist for their clients.
- While agent A is typically focused on *what* should be achieved, agents B and, mostly, C, should also put emphasis on *how* their goal is reached.
- Hence, faithful knowledge transfer arises as a natural prerequisite for tasks where an agent's behavior does matter.

Optimization as a Search Problem

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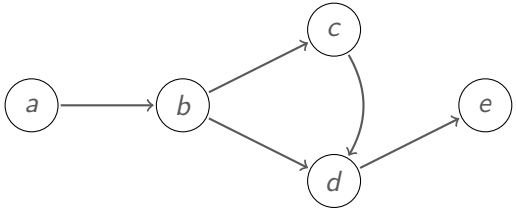
- Thus, the *what* aspect corresponds to the goal state, g, the agent has to reach.
- Similarly, the *why* aspect corresponds to the shape of the optimal path, p.
- Essentially, by applying restrictions on the shape of that path, the agent is searching for a tuple, (p, g), such that p connects its starting state, s, with the end state, g, while p adheres to a set of constraints, $\mathcal{C}$.

Argumentation Frameworks



$$\mathcal{A} = \{a, b, c, d, e\}$$
$$\mathcal{R} = \{(a, b), (b, c), (c, d), (b, d), (d, c), (d, e)\}$$

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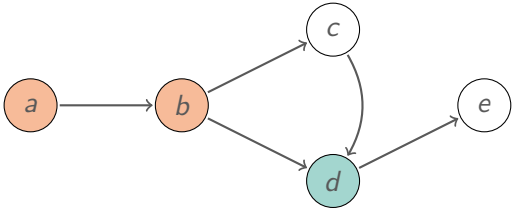

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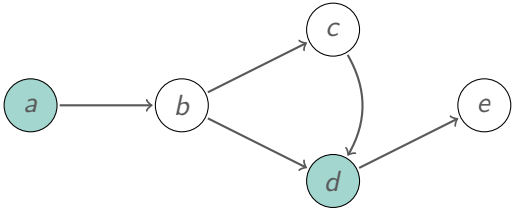

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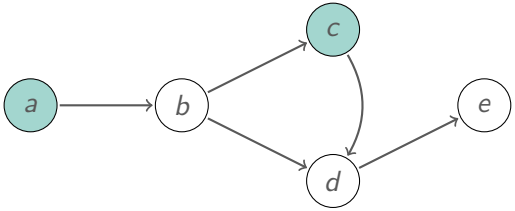

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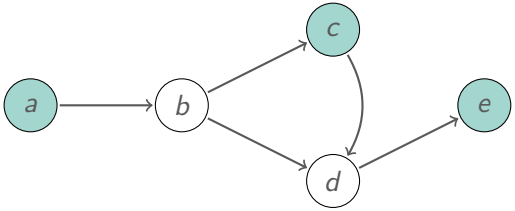

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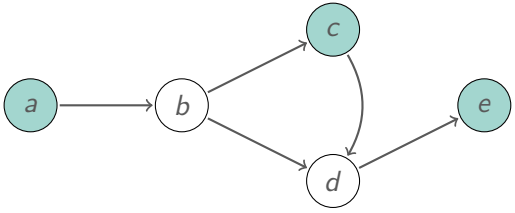

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- E is a **grounded** extension of $\langle \mathcal{A}, \mathcal{R} \rangle$ if it is the smallest complete extension of $\langle \mathcal{A}, \mathcal{R} \rangle$.

Machine Coaching

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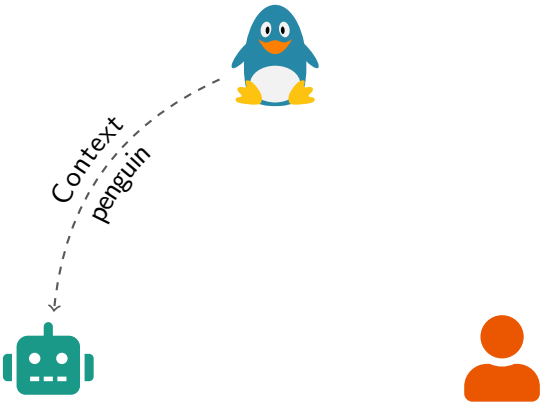

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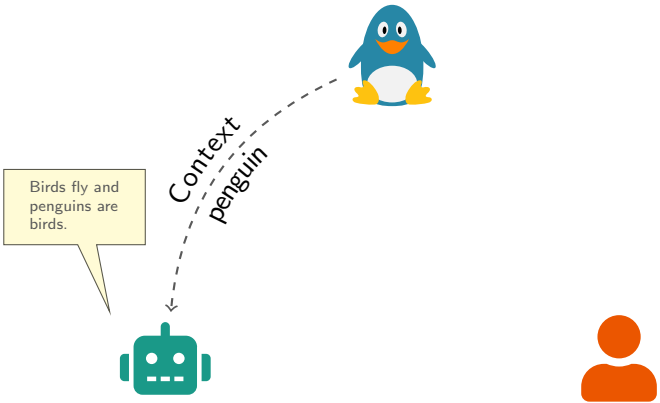

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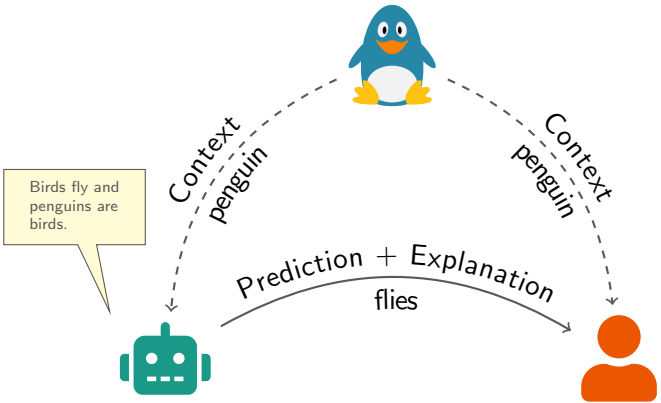

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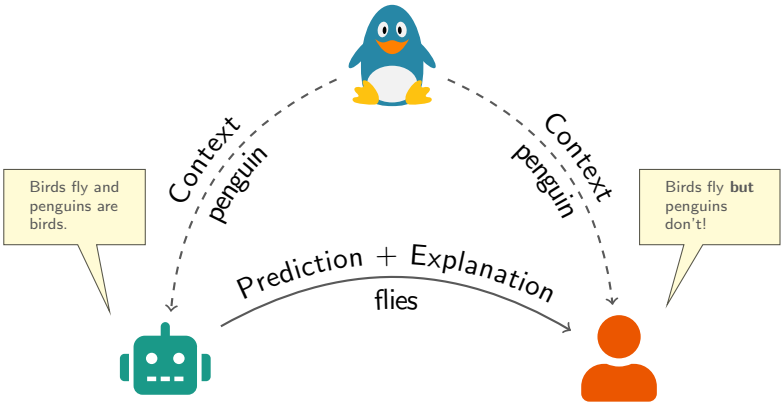

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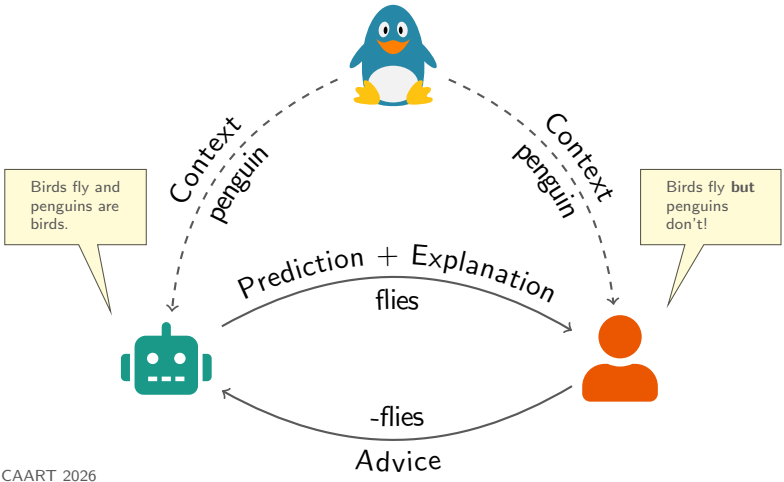

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- Empirically verified learning efficiency and efficacy.

Machine Coaching and AFs

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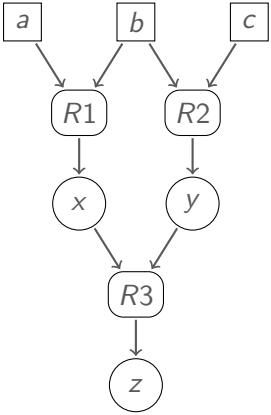

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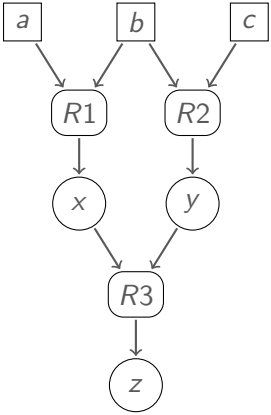

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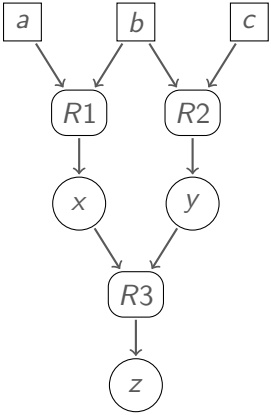

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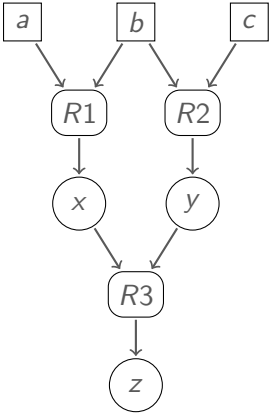

Reasoning in Machine Coaching

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a; b; c;

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R1 :: a implies x;
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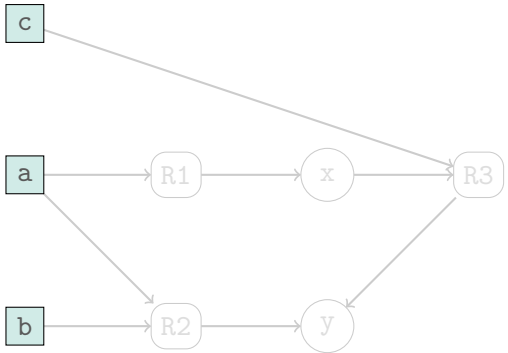

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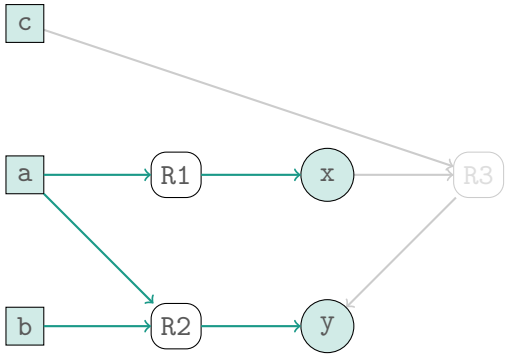

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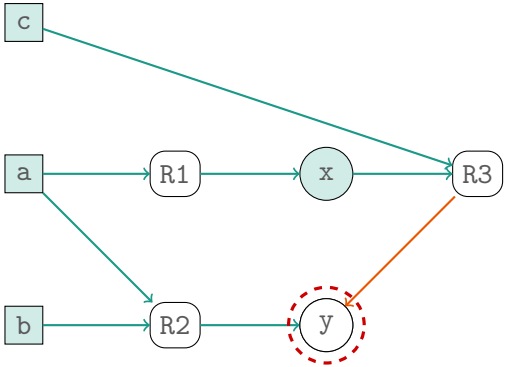

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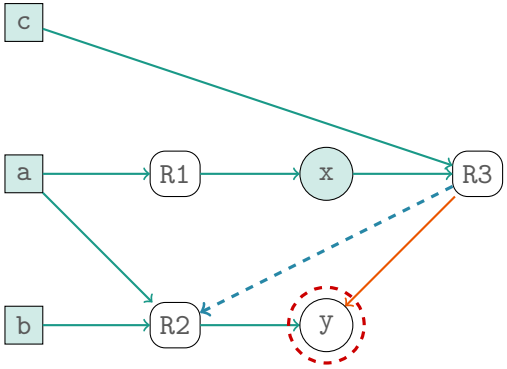

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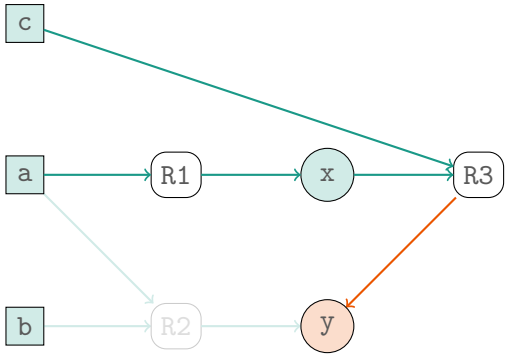

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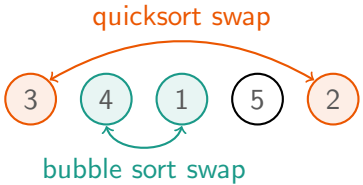
  - how efficient the learning process is.

Sorting Rules




```
Rbf :: (3,4,1,5,2) implies swap(4,1);
Rbp :: (_,4,1,_,_) implies swap(4,1);
Rqf :: (3,4,1,5,2) implies swap(3,2);
Rqp :: (3,_,_,_,2) implies swap(3,2);
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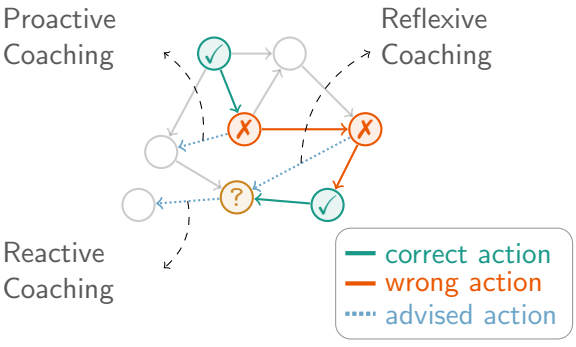
Coaching Styles



Coaching Process

Data: s, g, *learner*, *coach*
Result: *learner* with updated search policy; *path* from s to g.
do
    $path \leftarrow learner.search\_path(s, g)$;
    $advice \leftarrow coach.evaluate(path, s, g)$;
    $learner.update\_hypothesis(advice)$;
while *advice is not* `null`;

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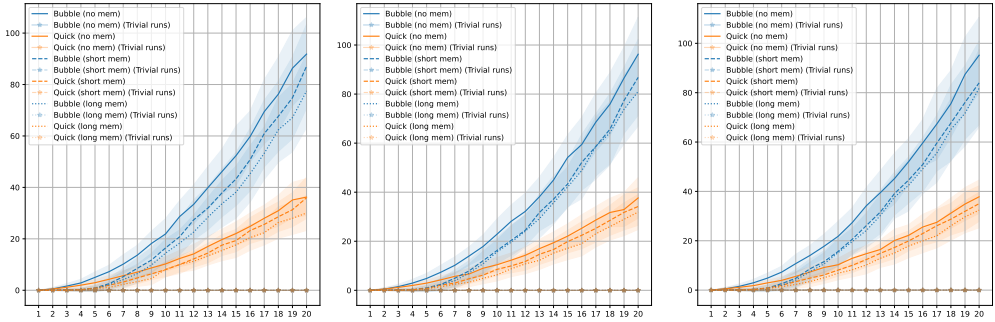
- *Coaching conformity* is the extent to which the agent's optimal path p corresponds to what the coaching policy would yield as an optimal path within a coaching session.

Experiments Run

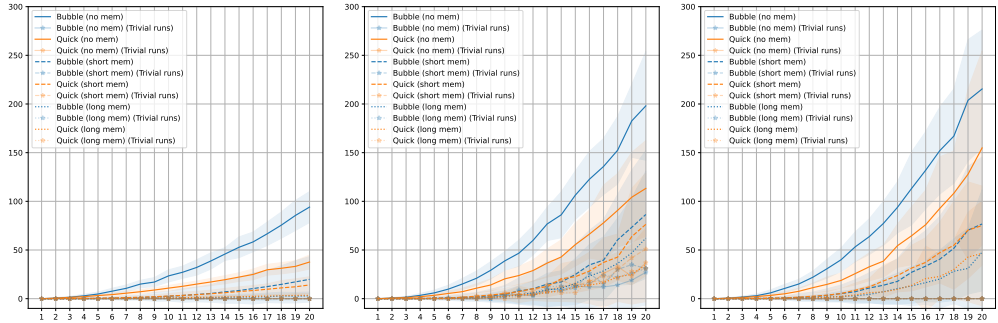
Attribute	Series A Experiments	Series B Experiments
State size, n	$\{1, 2, \ldots, 20\}$	$\{5, 10, 15, 20\}$
Iterations, m	100	20
Advice type	Full / Partial	partial@k
Memory	long / short / no	
Coaching style	reactive / reflexive / proactive	
Coaching policy	bubble sort, quicksort	

Series A: Full Advice Coaching Steps



(a) Reactive (b) Reflexive (c) Proactive

Series A: Partial Advice Coaching Steps



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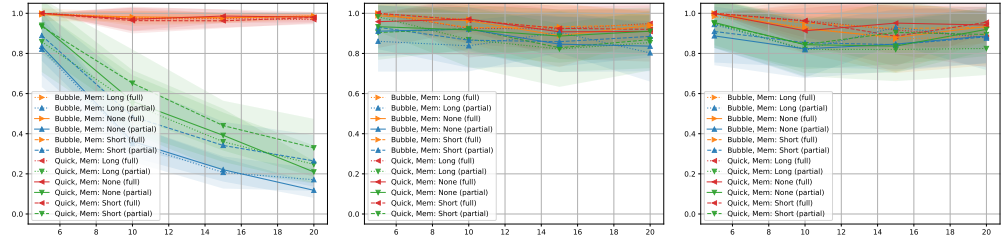
Series B: Coaching Conformity vs State Size, n



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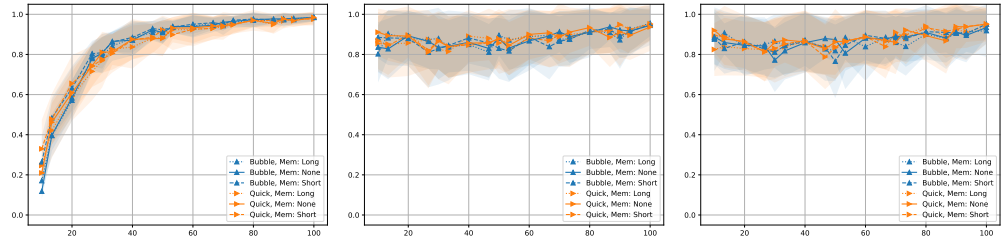
Series B: Coaching Conformity vs Advice Coverage



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- Next steps in this exploration might include but not restrict to:

Discussion & Future Work

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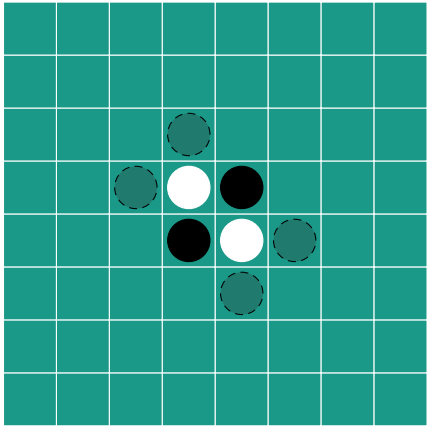
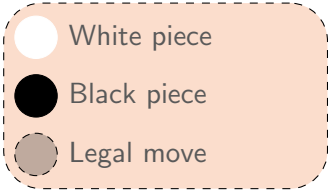
  - application in realistic settings.

Appendix

How to play Othello?

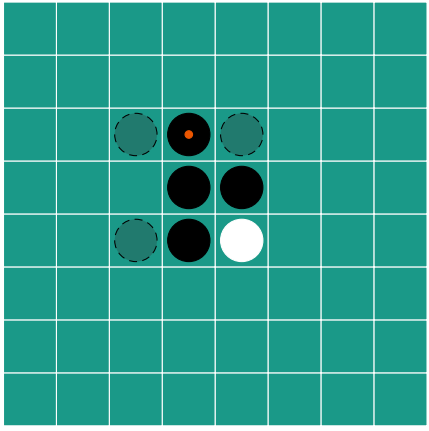




Rules

- Black plays first.
- Any new piece may be placed so as to enclose consecutive opponent pieces within friendly ones on the same row/column/diagonal.
- Any enclosed pieces are "converted" to the opponent's color.
- The player with the most pieces once the board is full or no more legal moves are allowed for any side wins.

How to play Othello?



White piece

Black piece

Legal move

Rules

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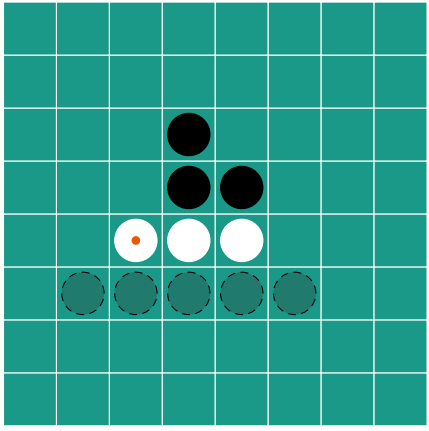
How to play Othello?



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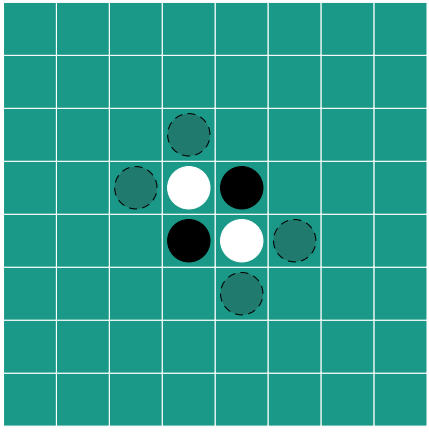
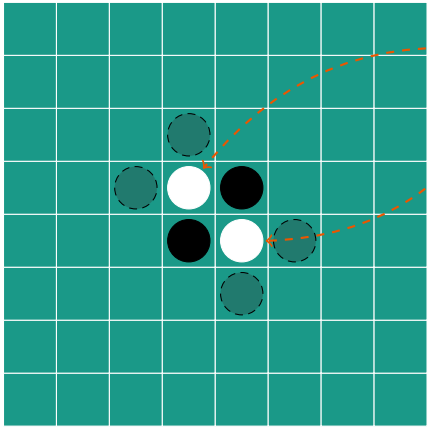
Othello's Language



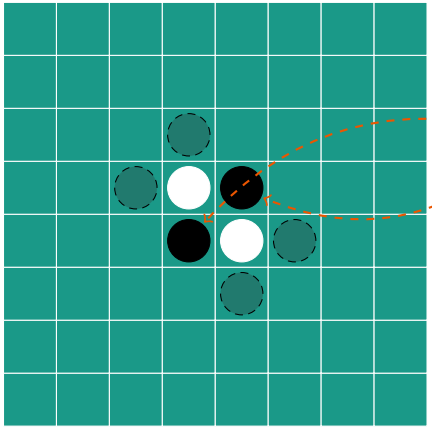
```
Context
cell(3, 3, 1);
cell(4, 4, 1);
cell(3, 4, -1);
cell(4, 3, -1);
legalMove(2, 3);
legalMove(3, 2);
legalMove(4, 5);
legalMove(5, 4);
cell(0, 0, 0); ...; cell(7, 7, 0);
```

Othello's Language



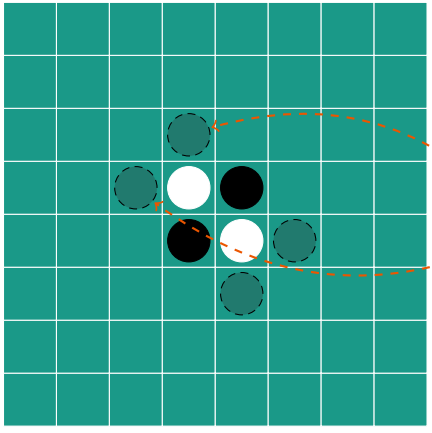
```
Context
cell(3, 3, 1);
cell(4, 4, 1);
cell(3, 4, -1);
cell(4, 3, -1);
legalMove(2, 3);
legalMove(3, 2);
legalMove(4, 5);
legalMove(5, 4);
cell(0, 0, 0); ...; cell(7, 7, 0);
```

Othello's Language



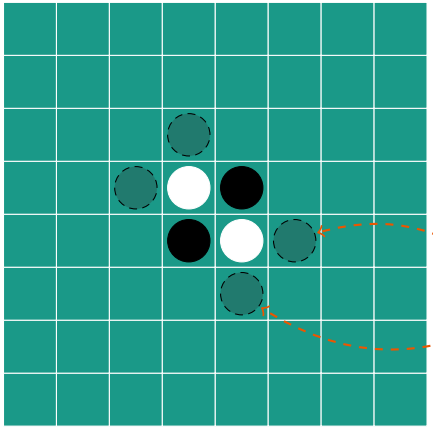
```
Context
cell(3, 3, 1);
cell(4, 4, 1);
cell(3, 4, -1);
cell(4, 3, -1);
legalMove(2, 3);
legalMove(3, 2);
legalMove(4, 5);
legalMove(5, 4);
cell(0, 0, 0); ...; cell(7, 7, 0);
```

Othello's Language



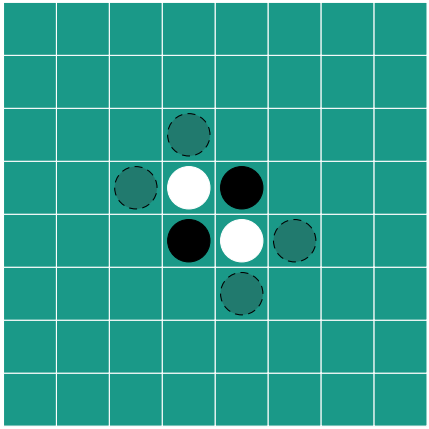
```
Context
cell(3, 3, 1);
cell(4, 4, 1);
cell(3, 4, -1);
cell(4, 3, -1);
legalMove(2, 3);
legalMove(3, 2);
legalMove(4, 5);
legalMove(5, 4);
cell(0, 0, 0); ...; cell(7, 7, 0);
```

Othello's Language



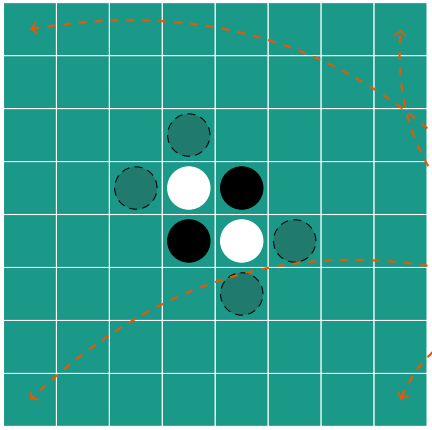
```
Context
cell(3, 3, 1);
cell(4, 4, 1);
cell(3, 4, -1);
cell(4, 3, -1);
legalMove(2, 3);
legalMove(3, 2);
legalMove(4, 5);
legalMove(5, 4);
cell(0, 0, 0); ...; cell(7, 7, 0);
```

A Dummy Policy



```
@Knowledge
D1 :: legalMove(X,Y) implies move(X,Y);
```

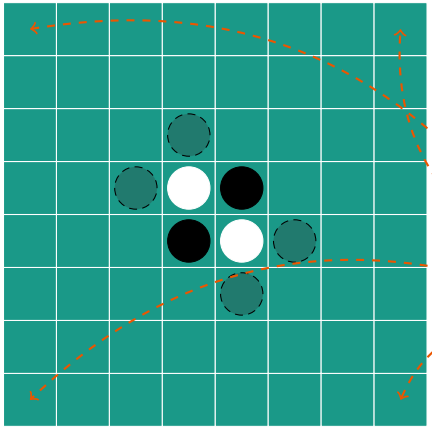
Simple Heuristics



```
@Knowledge
D1 :: legalMove(X,Y) implies move(X,Y);
D2 :: legalMove(X,Y) implies -move(X,Y);

R1 :: legalMove(0,0) implies move(0,0);
R2 :: legalMove(0,7) implies move(0,7);
R3 :: legalMove(7,0) implies move(7,0);
R4 :: legalMove(7,7) implies move(7,7);
```

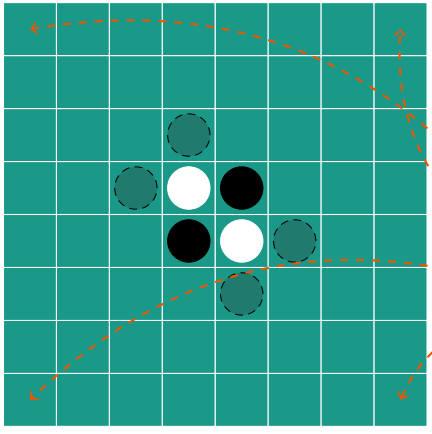
Simple Heuristics



```
@Knowledge
D1 :: legalMove(X,Y) implies move(X,Y);
D2 :: legalMove(X,Y) implies -move(X,Y);

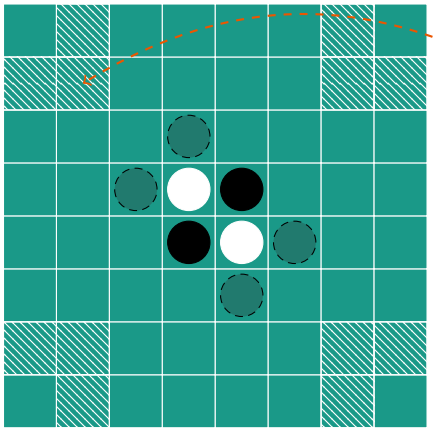
C1 :: implies corner(0,0);
C2 :: implies corner(0,7);
C3 :: implies corner(7,0);
C4 :: implies corner(7,7);
```

Simple Heuristics



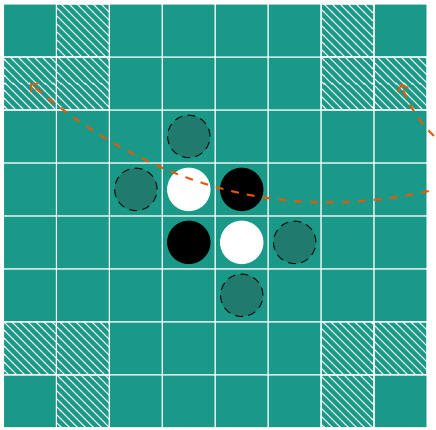
```
@Knowledge
D1 :: legalMove(X,Y) implies move(X,Y);
D2 :: legalMove(X,Y), legalMove(Z,W),
      corner(Z,W), -?=(X,Z), -?=(Y,W)
      implies -move(X,Y);
C1 :: implies corner(0,0);
C2 :: implies corner(0,7);
C3 :: implies corner(7,0);
C4 :: implies corner(7,7);
R1 :: legalMove(X,Y), corner(X,Y) implies
      move(X,Y);
```

More Complex Heuristics



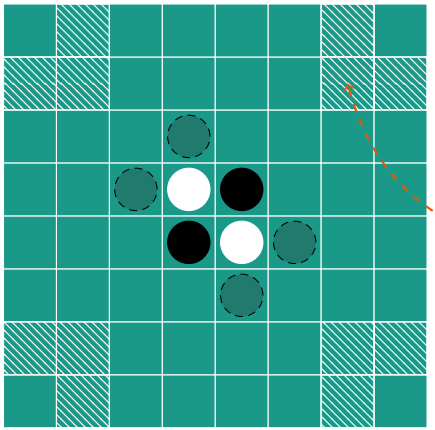
```
@Knowledge
...
A1 :: corner(X,Y), ?=(Z,X+1), ?=(W,Y+1)
      implies adjC(Z,W);
```

More Complex Heuristics



```
@Knowledge
...
A1 :: corner(X,Y), ?=(Z,X+1), ?=(W,Y+1)
      implies adjC(Z,W);
A2 :: corner(X,Y), ?=(Z,X+1), ?=(W,Y)
      implies adjC(Z,W);
```

More Complex Heuristics



```
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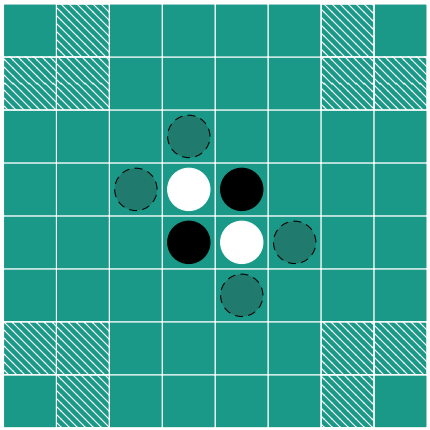
...
A1 :: corner(X,Y), ?=(Z,X+1), ?=(W,Y+1)
      implies adjC(Z,W);
A2 :: corner(X,Y), ?=(Z,X+1), ?=(W,Y)
      implies adjC(Z,W);
A3 :: corner(X,Y), ?=(Z,X+1), ?=(W,Y-1)
      implies adjC(Z,W);
```

More Complex Heuristics



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      implies adjC(Z,W);
A2 :: corner(X,Y), ?=(Z,X+1), ?=(W,Y)
      implies adjC(Z,W);
A3 :: corner(X,Y), ?=(Z,X+1), ?=(W,Y-1)
      implies adjC(Z,W);
...
```
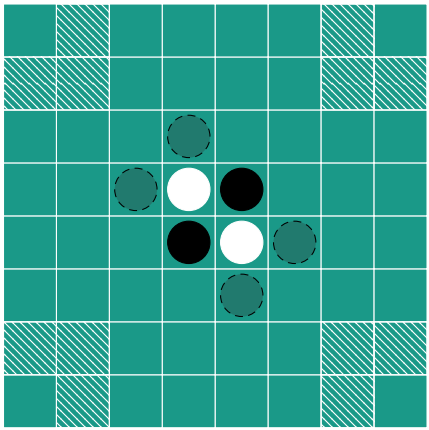
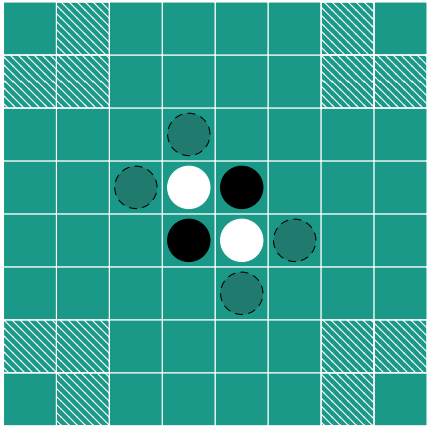

More Complex Heuristics



```
@Knowledge
...
@Procedures
function isAdj(x,y,z,w) {
    const X = parseInt(x);
    const Y = parseInt(y);
    const Z = parseInt(z);
    const W = parseInt(w);
    return Z-X < 2 && X-Z < 2 && W-Y < 2 &&
    Y-W < 2 && (X != Z || Y != W);
}
```

More Complex Heuristics



```
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    const Y = parseInt(y);
    const Z = parseInt(z);
    const W = parseInt(w);
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    Y-W < 2 && (X != Z || Y != W);
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```
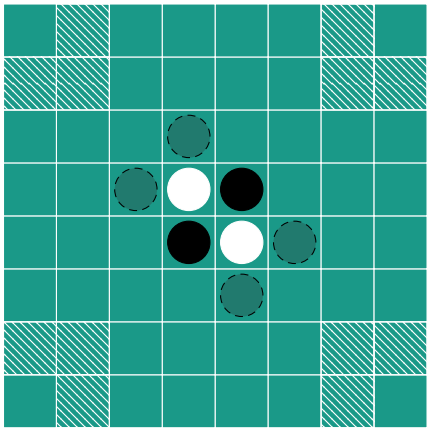
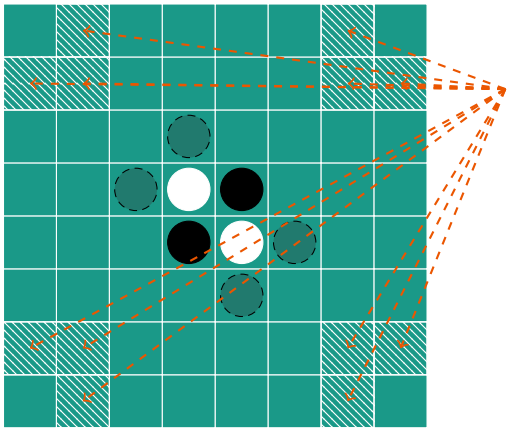

JavaScript

More Complex Heuristics



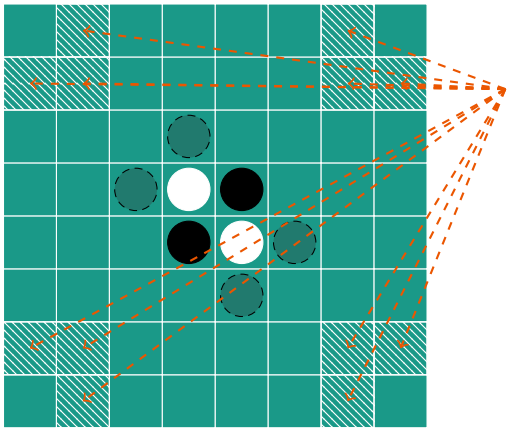
```
@Knowledge
...
R2 :: corner(X,Y), legalMove(Z,W),
      ?isAdj(X,Y,Z,W) implies -move(Z,W);
...
@Procedures
...
```

More Complex Heuristics



```
@Knowledge
...
R2 :: corner(X,Y), legalMove(Z,W),
      ?isAdj(X,Y,Z,W) implies -move(Z,W);
...
@Procedures
...
```

More Complex Heuristics



```
@Knowledge
...
R2 :: corner(X,Y), legalMove(Z,W),
      ?isAdj(X,Y,Z,W) implies -move(Z,W);
...
@Procedures
...
```

Benefits:

- More compact representation.
- Faster compilation.

Thank You!

Any Questions?